Robust Performance Synthesis

Understanding what D–K iteration actually changes

A controller can be excellent for the nominal model and poor for a slightly different plant. Chapter 6 gave us a way to **test** that risk; chapter 7 gave us a way to **design** for an ordinary H_∞ bound. This chapter explains how to use the second tool to improve the first test.

There are only two moving parts. **The controller K changes the physical closed loop. The scaling D changes how we certify that loop.** Confusing these roles makes the algorithm look much more mysterious than it is.

> The route through this note is: define the loop being tested → understand why scaling preserves the uncertainty problem → alternate analysis and controller design → check what the final number actually proves.

1 First fix the signals, not the algorithm

Use the negative-feedback tracking convention throughout the examples:

$$e = r - y, \quad u = Ke, \quad y = G(u + v), \quad q = Wu, \quad v = \Delta q. \tag{1}$$

Thus the true plant from u to y is $G(I + \Delta W)$. Here G is the nominal plant, W is a known uncertainty weight, and Δ is the unknown normalized perturbation. The performance output is $z = W_p e$. A small z means the specified tracking error is small, not necessarily that every unweighted signal is small.

Where do the two loops go? Close the controller loop first, but leave the uncertainty ports v, q open. The remaining stable system is called M_K:

$$\begin{pmatrix} q \\ z \end{pmatrix} = M_K \begin{pmatrix} v \\ r \end{pmatrix}, \quad v = \Delta q. \tag{2}$$

Nominal internal stability is a prerequisite: an unstable M_K is not rescued by a small frequency-response plot.

For this tracking configuration, let $S = (I + GK)^{-1}$. Eliminating e gives

$$e = Sr - SGv, \quad u = KSr - KSGv, \tag{3}$$

and hence

$$M_K = \begin{pmatrix} -WKSG & WKS \\ -W_pSG & W_pS \end{pmatrix}. \tag{4}$$

The top row describes uncertainty feedback; the bottom row describes weighted error. Keeping the off-diagonal blocks matters: uncertainty changes the error through them.

2 What exactly are we trying to make smaller than one?

The design requirement has two parts: **every allowed plant is internally stable**, and **every allowed plant satisfies** $\|T_{zr}(\Delta)\|_\infty < 1$, where T_{zr} denotes the transfer from r to z. Equivalently, the weighted disturbance-to-error energy gain must be below one uniformly over the uncertainty family.

The Main Loop Theorem packages these two requirements into one structured robustness test. Append a fictitious full block Δ_p from z back to r:

$$\Delta_e = \operatorname{diag}(\Delta, \Delta_p). \tag{5}$$

The appended block is a **test device**, not an additional physical modelling error. A performance channel with gain larger than one can be destabilized by an appropriately chosen full block of norm below one. This is why performance can be tested as stability of an extended loop.

For the standard complex, norm-bounded LTI uncertainty setting of chapters 4 and 6, the target, with a strict margin, is

$$\gamma_{\mu(K)} := \sup_\omega \mu_{\Delta_e}\left(M_{K(i\omega)}\right) < 1. \tag{6}$$

Real constant parameter blocks require extra care; the complex-block test used below can conservatively overbound them. Frequencies include zero and the limit at infinity.

2.1 An exact criterion is not yet a convenient design method

For a matrix M, $\mu_{\Delta_e}(M)$ is the reciprocal of the smallest structured perturbation norm that makes $I - M\Delta_e$ singular (and is zero if no such perturbation exists). It is **not defined as an optimization over scalings**. Scalings give an upper bound on it.

Directly optimizing $\gamma_{\mu(K)}$ is difficult: the destabilizing perturbation is structured, the worst frequency is unknown, and the closed-loop matrix depends nonlinearly on the controller. There is no general-purpose global solution supplied by the chapter. This does not mean μ is never exactly computable; special structures have exact formulas. It means that the general structured synthesis problem is not reduced to the ordinary H_∞ machinery of chapter 7.

Instead, design for the computable sufficient condition

$$\mu_{\Delta_e}\left(M_{K(i\omega)}\right) \le \inf_{D\in\mathcal{D}} \left\|D^{-1} M_{K(i\omega)} D\right\|_2. \tag{7}$$

The subscript 2 here means the largest singular value of a **matrix at one frequency**. The H_∞ norm is the supremum of that value over frequency for a stable system.

> An upper bound below one certifies success. An upper bound above one means **this certificate did not succeed**. It is not a proof of robust-performance failure, and certainly not a proof that no controller can succeed.

3 Why is changing D legitimate?

Suppose the uncertainty has two independent scalar channels and a full performance block. An admissible scaling is

$$D = \operatorname{diag}\big(d_1, d_2, d_p I\big), \quad d_1, d_2, d_p > 0. \tag{8}$$

It commutes with every allowed Δ_e. Therefore

$$D^{-1}(I - M\Delta_e)D = I - (D^{-1}MD)\Delta_e, \tag{9}$$

so the scaled and original interconnections become singular for exactly the same uncertainties. Small gain applied to the scaled matrix gives the bound in Equation 7.

In signal terms, define $\tilde{q} = D^{-1}q$ and $\tilde{v} = D^{-1}v$ on the extended ports. The uncertainty relation remains $\tilde{v} = \Delta_e \tilde{q}$. We changed the rulers used to measure internal signals, not the plant, the uncertainty budget, or the physical performance weight W_p.

3.1 A two-by-two calculation worth doing once

Take a fixed-frequency matrix and two independent complex scalar uncertainties:

$$M = \begin{pmatrix} 0 & 10 \\ 0.01 & 0 \end{pmatrix}, \quad \Delta_e = \operatorname{diag}(\delta_1, \delta_2). \tag{10}$$

Its unscaled norm is 10, so unscaled small gain says nothing useful. But

$$\det(I - M\Delta_e) = 1 - 0.1\delta_1\delta_2. \tag{11}$$

Destabilization requires $\delta_1\delta_2 = 10$. The smallest possible $\max(|\delta_1|, |\delta_2|)$ is $\sqrt{10}$, so $\mu(M) = \frac{1}{\sqrt{10}} \approx 0.3162$.

Now set $D = \operatorname{diag}(d, 1)$. Then

$$D^{-1}MD = \begin{pmatrix} 0 & \frac{10}{d} \\ 0.01d & 0 \end{pmatrix}, \tag{12}$$

whose largest singular value is $\max\big(\frac{10}{d}, 0.01d\big)$. Balance the two terms:

$$\frac{10}{d} = 0.01d \quad \Longrightarrow \quad d = \sqrt{1000}, \quad \big\|D^{-1}MD\big\|_2 = \sqrt{0.1}. \tag{13}$$

The scaling recovers the exact answer in this example. **Nothing physical improved**: the matrix was already robust, but the original norm obscured that fact.

3.2 Which scalings are allowed?

- An independent **full** uncertainty block allows a scalar multiple of the identity.
- A **repeated scalar** block δI allows a matrix scaling within that repeated block.
- Blocks must retain their structure; an arbitrary mixing of channels is not allowed.

A common scalar multiplier on all of D cancels from $D^{-1}MD$. Consequently the performance-block scaling can be normalized to I. For the example above, our $\operatorname{diag}\big(d_1, d_2, d_p, d_p\big)$ is equivalently $\operatorname{diag}\Big(\frac{d_1}{d_p}, \frac{d_2}{d_p}, 1, 1\Big)$.

4 The algorithm: one variable fixed at a time

Write the joint surrogate as

$$J(K, D) = \sup_{\omega} \left\| D(\omega)^{-1} M_{K(i\omega)} D(\omega) \right\|_2. \tag{14}$$

There are two different jobs hiding in this expression.

4.1 D-step: analyze the controller you already have

Hold $K = K_j$ fixed. At each frequency, search the admissible scalings for a small largest singular value. Record both the bound and a scaling that achieves it. This step changes **only the certificate**. A better D-step does not change a step response. One can formulate a fixed-frequency bound test using positive definite commuting matrices and an LMI; a grid search alone is not an exact optimizer.

4.2 Fit: turn frequency data into a realizable weight

The samples $D(\omega_k)$ are not yet a transfer matrix. To use ordinary H_∞ synthesis, construct a structured real-rational $\hat{D}(s)$ that is proper and stable, with a proper stable inverse. For independent scalar blocks, one commonly fits stable minimum-phase scalar weights to the required magnitude profiles. Their phase is not an extra physical uncertainty: diagonal unit-modulus phases give unitary changes of coordinates and leave singular values unchanged.

For example, $\hat{d}(s) = \frac{c(s+a)}{s+b}$ with $a, b, c > 0$ has a stable pole, a stable zero, and finite nonzero limits at zero and infinity. Both it and its inverse are proper and stable. A constant positive d is the simplest possible fit, of order zero.

The important check is the **scaled bound after fitting**, not just whether a curve looks smooth. A small absolute fitting error can matter greatly if a scaling or its inverse is poorly conditioned.

4.3 K-step: design a new physical controller

Let $\mathcal{K}$ denote the internally stabilizing controllers. Hold $\hat{D}$ fixed and solve

$$\inf_{K \in \mathcal{K}} \left\| \hat{D}^{-1} M_K \hat{D} \right\|_\infty. \tag{15}$$

This is a standard weighted H_∞ problem. Scale only the external uncertainty and performance ports of the generalized plant; keep the measurement and control ports in their original units. The resulting controller acts on the **original** physical loop.

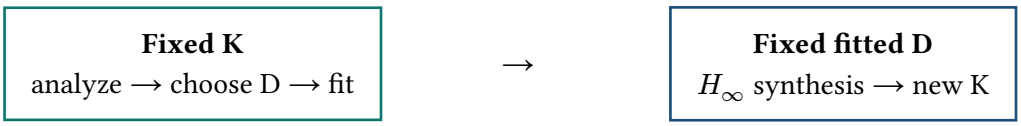


Figure 1: Repeat with the new controller. Re-analyze the original uncertainty structure after each physical change; stop at a verified bound below one or at stagnation.

5 What decreases, and what does not follow?

Imagine exact alternating minimization in one common admissible class, with no fitting error. Since the old value remains feasible at each update,

$$J(K_j, D_{j+1}) \le J(K_j, D_j), \tag{16}$$

$$J(K_{j+1}, D_{j+1}) \le J(K_j, D_{j+1}). \tag{17}$$

These inequalities are **non-strict**. If the current controller and scaling are already best responses to each other, neither step improves anything.

The nonnegative objective values then converge. That does not prove convergence of the controllers, convergence of the scalings, or local or global optimality. The joint problem is not made convex by solving its two easier subproblems in turn.

In a practical frequency-dependent implementation, fitting changes the object between the two steps. Approximate synthesis, finite frequency grids, and later controller reduction introduce further discrepancies. Recompute the final bound rather than assuming ideal monotonicity survived these operations.

Observation	What it licenses you to say
Stable scaled loop, H_∞ bound < 1	Robust performance is certified for the stated uncertainty model.
Sampled bound < 1	The sampled frequencies pass; intersample peaks still need checking.
D-upper-bound > 1	No certificate from that bound. Failure remains unproved.
Admissible uncertainty produces a weighted gain > 1	A counterexample proves failure for that controller.
Iteration stagnates	This run stopped improving; feasibility or optimality is not settled.

5.1 Why not fit a very high-order D?

A dynamic fit enlarges the weighted synthesis plant, because both $\hat{D}$ and $\hat{D}^{-1}$ enter. Standard full-order synthesis can therefore produce a controller with order on the scale of $\operatorname{order}(P) + 2\operatorname{order}\left(\hat{D}\right)$ before cancellations and reduction. This is an order-cost warning, not a universal equality for a minimal controller. If the controller is reduced afterwards, the reduced controller must be checked again against the original uncertainty and performance requirements.

5.2 Why chapter 9 will not make D–K iteration obsolete

Youla coordinates can make M_K affine in a stable parameter Q. For a **fixed** scaling, a norm objective is then convex in Q. But optimizing the similarity $D^{-1}M(Q)D$ jointly over Q and D still couples the variables nonlinearly. Chapter 9 removes the stabilization constraint from the design coordinates; it does not remove the structured-robustness difficulty.

6 Worked exercise: why exact decoupling is fragile

This is the exercise on pp. 222–223 of [1]. All times below are in **minutes**, all frequencies in radians per minute. The plant and weights are

$$G(s) = \frac{G_0}{75s+1}, \quad G_0 = \begin{pmatrix} 87.8 & -86.4 \\ 108.2 & -109.6 \end{pmatrix}, \tag{18}$$

$$W(s) = \frac{s+0.1}{0.5s+1} I, \quad \Delta = \mathrm{diag}(\Delta_1, \Delta_2), \quad \|\Delta\|_\infty < 1. \tag{19}$$

The two channels have independent stable, real-rational dynamic perturbations. They are not required to be real-valued at each frequency. A constant real diagonal perturbation is one allowed special case.

6.1 Part (a): interpret the uncertainty

$$|w(0)| = 0.1, \quad |w(i)| \approx 0.899, \quad |w(10i)| \approx 1.961, \quad |w(\infty)| = 2. \tag{20}$$

Thus the input-channel relative error is small at low frequency and can be very large at high frequency. A bound greater than 100% does not mean literally no model information; it means the allowed relative discrepancy exceeds the nominal magnitude. Diagonal uncertainty changes the two input channels independently; it does not allow an arbitrary cross-channel perturbation.

6.2 Part (b): start with the tempting controller

The exercise proposes $K_\beta = \left(\frac{\beta}{s}\right) G^{-1}$ for $\beta \in \{0.1, 0.3, 0.5, 0.7\}$. Since $GK_\beta = K_\beta G = \left(\frac{\beta}{s}\right) I$,

$$S = \frac{s}{s+\beta} I, \quad T = I - S = \frac{\beta}{s+\beta} I. \tag{21}$$

For a simultaneous unit step in both references, the nominal error in each output is $e^{-\beta t}$. The 2% settling time is $-\frac{\ln(0.02)}{\beta} \approx \frac{3.912}{\beta}$: about $39.1, 13.0, 7.82, 5.59$ minutes. Faster nominal response looks free.

Choose the exercise's weight

$$W_{p(s)} = \alpha \frac{s+\beta}{s+10^{-6}} I. \tag{22}$$

Then $W_p S = \alpha \frac{s}{s+10^{-6}} I$, so the nominal performance norm is exactly α. At $\alpha = 1$ it reaches the non-strict boundary; it has no strict margin.

The uncertainty-return matrix is $-WK_\beta SG = -w\frac{\beta}{s+\beta} I$. Its structure is diagonal, so the robust-stability test reduces to the scalar norm:

β	0.1	0.3	0.5	0.7
$\left\| w\frac{\beta}{s+\beta} \right\|_\infty$	0.100	0.263	0.402	0.520

All four controllers are robustly stable. That still says nothing about whether the stable perturbed loop tracks well.

6.3 The mechanism: inversion amplifies unequal input errors

G_0 has condition number about 141.7. Its inverse creates large input combinations whose effects almost cancel. Unequal gain errors disturb that cancellation.

Take $\Delta = \mathrm{diag}(-0.99, 0)$, a constant real uncertainty with norm 0.99. At DC it reduces the first input-channel gain by $0.99 \times 0.1 = 9.9\%$. It is **not** a 1% gain error. At other frequencies its physical effect includes the dynamics of $w(s)$.

Let $h_j = 1 + \delta_j w(s)$ and $f_j = \frac{s}{s+\beta h_j}$. Direct substitution gives

$$S_\Delta = G_0 \ \mathrm{diag}(f_1, f_2) G_0^{-1}. \tag{23}$$

Nominally the two f_j are identical and the similarity cancels. With only the first input perturbed, $f_1 \neq f_2$, and the nearly singular G_0 matters again. For a reference step in direction $\begin{pmatrix}1\\1\end{pmatrix}$, the first error component combines the two scalar step-error responses with coefficients about 7.42 and −6.42. Their sum is one, but a small difference between the two responses is multiplied by a large number.

For $\beta = 0.5$, simulation gives

Quantity	Nominal	Bad uncertainty
Error at $t = 2$	$(0.368, 0.368)$	$(1.546, 1.820)$
Largest error, $0 \le t \le 60$	$(1, 1)$	$(3.230, 3.825)$
Smallest error, same window	approximately $(0, 0)$	$(-0.610, -0.775)$

The error should decay from one. Instead it first grows dramatically and then changes sign. The poles remain stable; this is bad performance, not instability.

6.4 Separate the picture from a proof of weighted-performance failure

A step plot is not an H_∞ test: an infinite-duration step is not an L_2 input. To disprove the specified weighted norm bound, inspect $W_p S_{\Delta(i\omega)}$ for the **same admissible uncertainty**. The script reports a frequency with largest singular value above one at $\alpha = 1$. That is an actual counterexample to the performance specification.

For comparison, numerical frequency-by-frequency D-upper-bound analysis of the nominal cut gives approximately $3.90, 6.19, 7.61, 8.65$ for the four gains at $\alpha = 1$. These sampled upper bounds alone would be inconclusive; the explicit uncertainty provides the missing failure evidence. The script prints both quantities separately.

7 Part (c): actually redesign the controller

To keep the computation transparent and reproducible, use the **restricted constant-D version** of the iteration. Constants are order-zero proper stable weights with proper stable inverses. This is a valid, more conservative subset of the dynamic-scaling method; it is not a frequency-dependent fit masquerading as one.

Fix $\beta = 0.5$ in W_p and relax α to 0.1. This makes the allowed weighted-error envelope ten times looser than at $\alpha = 1$. We are not claiming that the original specification has been achieved.

7.1 Assemble the synthesis plant from the signal equations

With rows ordered as (q, z, e) and columns as (v, r, u), the generalized plant is

$$P = \begin{pmatrix} 0 & 0 & W \\ -W_p G & W_p & -W_p G \\ -G & I & -G \end{pmatrix}, \quad u = Ke. \tag{24}$$

Closing K gives exactly Equation 4. No negative sign is added to K: the negative feedback already appears in the last row of P.

Start with $K_0 = K_{0.5}$. At each of two iterations:

1. Choose $D = \mathrm{diag}(d_1, d_2, 1, 1)$ to reduce the worst scaled singular value of the current loop on a frequency grid. Search over $\log(d_1), \log(d_2)$ to preserve positivity.
2. Form $P_D = \mathrm{diag}(D^{-1}, I) P \ \mathrm{diag}(D, I)$ and run Octave's `hinfsyn(P_D,2,2)`.
3. Independently evaluate the continuous-frequency H_∞ norm of $D^{-1} M_K D$ and check nominal closed-loop poles. Do not turn the grid maximum into an all-frequency claim.

The first scaling is approximately $\mathrm{diag}(1.655, 1.652, 1, 1)$. Its initial bound is about 32.60: a **constant** scaling is a poor certificate for the inversion controller over the entire frequency band. The controller update is the important improvement: it produces a sixth-order controller with continuous-frequency scaled norm about **0.7911**, below one. The second pair of updates stagnates at essentially the same value; solver-level differences in the sixth decimal are not meaningful improvement. The script retains the incumbent if a candidate is worse under the new scaling.

This is a robust-performance certificate for the relaxed specification and the entire stated diagonal uncertainty family, subject to numerical solver accuracy. It is not merely a successful test against the selected bad uncertainty. Since the bound is below one even with constant D, no dynamic fit is necessary to certify this particular design.

7.2 Compare what changed physically

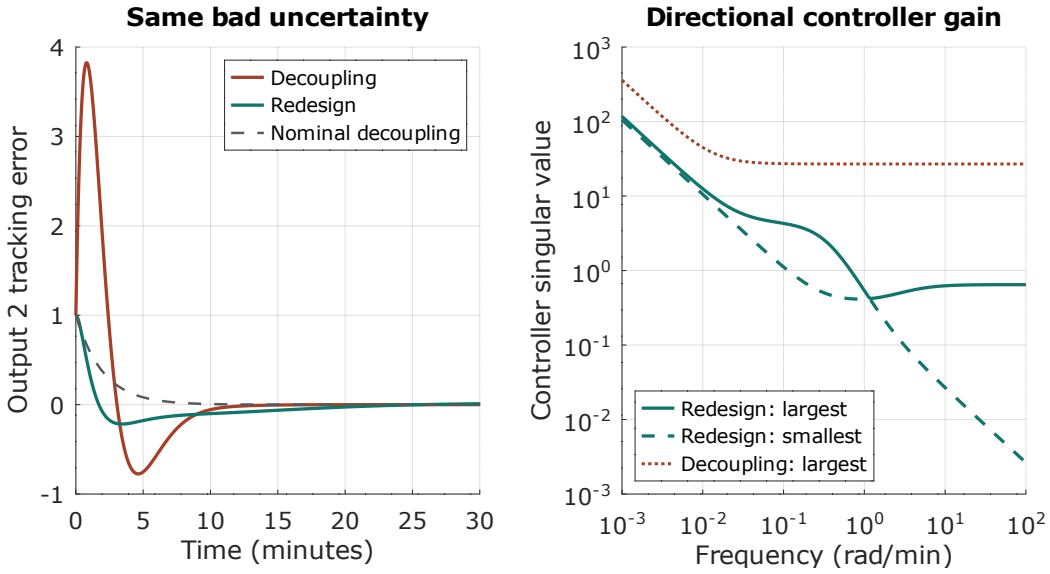


Figure 2: Computed, not sketched. Left: error in output 2 for a simultaneous unit reference step under $\Delta = \mathrm{diag}(-0.99, 0)$. Right: the two singular-value curves of the redesigned controller; the largest one sets the worst-direction gain. Nominal decoupling has large gain in the weak plant direction.

For the same bad uncertainty, the redesigned errors have maxima about $(1, 1)$ and minima about $(-0.017, -0.217)$ over 60 minutes. At two minutes they are about $(0.132, -0.086)$ instead of $(1.546, 1.820)$. Thus the dramatic initial growth is gone, although output 2 still changes sign. Robust performance is not a promise of monotone step responses.

The frequency plot is as important as the step plot: the redesigned controller no longer insists on the exact inverse gain in every plant direction. It trades the nominal decoupling recipe for a worst-case weighted design. The selected step is an illustration; the scaled H_∞ bound is the family-wide evidence.

7.3 Reproduce the exercise

From the repository root, with Octave's installed `control` package:

```
octave --no-gui --quiet \
  Control/robust-control-08-robust-performance-synthesis/scripts/distillation.m
```

The script reproduces all reported numbers, pole checks, and this figure. Its finite-grid D-search is not claimed to be globally optimal.

8 Keep these four statements distinct

1. **K changes the loop; D changes the certificate.**
2. **A successful upper bound proves success; an unsuccessful upper bound does not prove failure.**
3. **Ideal alternating minimization is non-increasing; practical fitting and numerical synthesis must be checked, and stagnation is not optimality.**
4. **Relaxing a weight changes the question being answered.** Always report the weight together with the final robustness number.

Source. Based on chapter 8, pp. 217–223 of [1]. The examples and computations are developed here. Dynamic D-fitting is explained, not implemented; the executable exercise uses constant scalings.

[1] C. W. Scherer, "Theory of Robust Control." [Online]. Available: https://www.imng.uni-stuttgart.de/mst/files/RC.pdf